**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

UNIVERSITY COLLEGE OF ENGINEERING (A)

Osmania University, Hyderabad-07

**Class:** B.E. (CSE / AIML)
**Semester:** V
**Subject:** Computer Vision

**Max Marks:** 30
**Duration:** 1 ½ hrs (9:30 a.m. – 11:00 a.m.)
**Date:** 23-09-2025

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## PART – A

**Answer ALL Questions (5 × 2 = 10 Mark)**

**Q1.**

Perform a **1D convolution** on the signal

$$x = [1,\ 2,\ 0,\ 3]$$

using the kernel

$$h = [1,\ -1]$$

with **stride = 1** and **no padding.**
Show all the steps of computation clearly.

b) Perform a 2D convolution on the input matrix

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

using the filter

$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

with **stride = 1** and **no padding.**

3) Convert the RGB pixel value (128,64,32) into **HSV** color space.
4) A pixel has RGB = (90, 150, 200). Convert it into **Lab** representation.
5) Differentiate between:
   a. Image normalization vs. standardization.
   b. Blurring filter vs. edge-detection filter .

## PART – B

**Answer ANY TWO Questions ( 2 × 10 = 20 Marks)**

6.

a) Derive the **pinhole camera model** equation and explain its limitations.
b) Explain perspective projection with a neat diagram.

7. Give the model summary for the following:

**CNN Architecture**

- Input layer: $64 \times 64 \times 3$
- Convolutional layer: 32 filters, $3 \times 3$ kernel, ReLU activation
- Max Pooling layer: $2 \times 2$ pool size
- Convolutional layer: 64 filters, $3 \times 3$ kernel, ReLU activation
- Max Pooling layer: $2 \times 2$ pool size
- Convolutional layer: 128 filters, $3 \times 3$ kernel, ReLU activation
- Max Pooling layer: $2 \times 2$ pool size
- Flatten layer
- Fully connected layer: 128 neurons, ReLU activation
- Output layer: 10 neurons (for classification into 10 classes), softmax activation

8.

a) Write the **OpenCV function** to rotate an image by a given angle $\theta$.
b) Derive the **mathematical formula (rotation matrix)** used internally with an example.